41. (c) Living cells shrinks in hypertonic solution (plasmolysis) while bursts in hypotonic solution (endosmosis). There is no. effect when living cells are kept in isotonic solution.

42. (b) Ionic concentration

Explanation:

Osmotic pressure depends on the number of solute particles (ions or molecules) present in the solution.

$$\pi = CRT$$

where:

- π= osmotic pressure
- C= concentration of solute particles
- R= gas constant
- T= absolute temperature

Hence, osmotic pressure is directly proportional to ionic concentration.

43. (c) $\pi V = nRT$

$$\pi = \frac{w}{m}\frac{RT}{V} = \frac{10}{342} \times \frac{0.821 \times (273 + 69)}{0.1} = 8.21 \textit{ atm}.$$

44. (a) Fructose

Explanation:

Small molecules can diffuse through cell membranes more easily. Fructose is a small sugar molecule, whereas:

- glycogen is a large polysaccharide
- haemoglobin is a large protein
- catalase is an enzyme (protein)

Therefore, fructose can diffuse through the membrane.

45. (c) KNO_3 dissociates completely while CH_3COOH dissociates to a small extent. Hence, $P_1 > P_2$.

46. (c) Directly proportional to temperature on the Kelvin scale

Explanation:

Osmotic pressure is given by:

$$\pi = CRT$$

where:

- π= osmotic pressure
- C= concentration
- R= gas constant
- T= absolute temperature (Kelvin)

Hence, osmotic pressure is directly proportional to temperature on the Kelvin scale.

47. (b) $\pi V = nRT$

$\frac{500\, V_1}{105.3\, V_2} = \frac{nR \times 283}{nR \times 298}$; $\frac{V_1}{V_2} = \frac{1}{5}$ so $V_2 = 5V_1$

48. (a) There is no net movement of the solvent through the semipermeable membrane between two solution of equal concentration.

49. (a) Directly proportional to the concentration

Explanation:

Osmotic pressure is given by:

$$\pi = CRT$$

At constant temperature T,

$$\pi \propto C$$

Therefore, osmotic pressure is directly proportional to concentration.

50. (b) $\pi V = \frac{w}{m} RT$

$\therefore\ 6 \times 10^{-4} \times 1 = \frac{4}{m} \times 0.0821 \times 300$; $m = 1.64 \times 10^{5}$.

51. (d) Osmosis

Explanation:

Osmosis is the movement of solvent molecules through a semipermeable membrane from a dilute solution to a concentrated solution.

52. (d) According to the dialysis process molecular weight increases but sensitivity decreases.

53 **Explanation:**

At $20°C$,

- Maximum NaCl dissolved in 100 g water $= 35\ g$
- Added NaCl $= 50\ g$

Undissolved NaCl:

$$50 - 35 = 15\ g$$

(a) 15 g

54. **(b) They have the same weight concentration**

Explanation:

Isotonic solutions only need equal osmotic pressure.

Their concentrations by weight may be different.

55. (d) $\pi \propto T$; if T is doubled π is also doubled.

56. (b) Osmosis reaction are takes place in increases the volume.

57. **(d) Increases or decreases**

Explanation:

During osmosis, solvent moves through a semipermeable membrane:

- into the solution if it is more concentrated → volume increases
- out of the solution if it is less concentrated → volume decreases

So the volume may either increase or decrease depending on the direction of solvent flow.

58. (a) For two non-electrolytic solution if isotonic, $C_1 = C_2$

$$\therefore \frac{8.6}{60 \times 1} = \frac{5 \times 1000}{m.wt. \times 100} \qquad \therefore m = 348.9$$

59. (b) Both urea and glucose are non-electrolytes but *NaCl* being electrolyte ionises.

60 **(b), (c) and (d)**

Explanation:

Osmotic pressure depends on total particle concentration:

$$\pi = iCRT$$

(a) 0.1 M NaCl

NaCl dissociates into 2 ions:

$$iC = 2 \times 0.1 = 0.2$$

(b) 0.1 M glucose

Glucose is non-electrolyte:

$$iC = 0.1$$

(c) 0.6 g urea in 100 mL

$$\text{Moles} = \frac{0.6}{60} = 0.01$$

Volume $= 0.1L$

$$M = \frac{0.01}{0.1} = 0.1M$$

Urea is non-electrolyte:

$$iC = 0.1$$

(d) 1.0 g solute X in 50 mL

$$\text{Moles} = \frac{1}{200} = 0.005$$

Volume $= 0.05L$

$$M = \frac{0.005}{0.05} = 0.1M$$

Non-electrolyte:

$$iC = 0.1$$

So (b), (c), and (d) have same osmotic pressure.

(b), (c) and (d)

61. **(a) and (c)**

Explanation:

(a) 0.01 M glucose

Non-electrolyte:

$$iC = 0.01$$

(b) 0.01 M $NaNO_3$

$$NaNO_3 \rightarrow Na^+ + NO_3^-$$
$$iC = 2 \times 0.01 = 0.02$$

(c) 0.3 g urea in 500 mL

$$\text{Moles} = \frac{0.3}{60} = 0.005$$

$$M = \frac{0.005}{0.5} = 0.01M$$

Non-electrolyte:

$$iC = 0.01$$

(d) 0.04 N HCl

For HCl:

$$N = M$$
$$iC = 2 \times 0.04 = 0.08$$

Hence (a) and (c) are isotonic.

(a) and (c)